

Weighted kappa measures for ordinal multi-class classification performance

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ABSTRACT

Assessing the classification performance of ordinal classifiers is a challenging problem under imbalanced data compositions. Considering the critical impact of the metrics on the choice of classifiers, employing a metric with the highest performance is crucial. Although Cohen's kappa measure is used for performance assessment, there are better-performing agreement measures under different formations of ordinal confusion matrices. This research implements weighted agreement measures as evaluation metrics for ordinal classifiers. The applicability of agreement and mainstream performance metrics to various practice fields under challenging data compositions is assessed. The sensitivity of the metrics in detecting subtle distinctions between ordinal classifiers is analyzed. Five kappa-like agreement measures with six weighting schemes are employed as evaluation metrics. Their reliability/usefulness is compared to the mainstream and recently proposed metrics, including F1, Matthews correlation coefficient, and informational agreement. The performance of 37 metrics is analyzed in two extensive numerical studies, including synthetic confusion matrices and real datasets. Promising metrics under practical circumstances are identified, and recommendations about the best metric to evaluate ordinal classifiers under different conditions are made. Overall, the weighted Scott's pi-measure is found useful, sensitive to small differences in the classification performance, and reliable under general conditions.

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1. Introduction

Classifying subjects into multiple ordinal classes, namely ordinal multi-class classification or ordinal classification, is one of the most frequent exercises of automatic classification systems in pattern recognition, machine learning, and deep learning systems. The problem considered here is to assign each object in a sample to one of the ordered classes of a categorical response variable using an ordinal classifier. In pattern recognition, ordinal multi-class classification is used in the classification of different types of images such as hyperspectral images [1], data-like images [2], radar images [3], or images from medical diagnosis systems [4]. Accurate evaluation of ordinal classifiers' performance is a challenge under different circumstances of data. Therefore, many metrics have been proposed to evaluate classifiers. The quality of training a model or network is related to the preciseness of the classifier against true classes in a labeled dataset [4]. A metric can also be used as a loss function to optimize in an image classification system [4]. In this sense, it is crucial to evaluate

the performance of ordinal classifiers with the highest possible accuracy.

The evaluation of ordinal classifiers' performance is directly related to the used evaluation metric and the characteristics of the training or test dataset, which we call "the composition of the dataset" throughout the manuscript. When the distribution of subjects into ordinal classes is imbalanced, mainstream metrics such as accuracy, precision, and recall are negatively impacted; hence, they do not precisely assess classifiers' performance [5,6]. Therefore, the use of other measures such as F1 score [7], Cohen's (weighted) κ (kappa)-measure [8] and Matthews correlation coefficient (MCC) [9] is proposed. MCC is observed to perform better than the F1 score and κ -measure for binary classification [10,11]. For multi-class classification, Rácz et al. [6] identify better performance for F1 score than MCC and find that MCC is more sensitive to the data composition. Although Rácz et al. [6] include Cohen's unweighted κ in their study, they do not give any specific inference about the comparison of Cohen's unweighted κ -measure to F1 score and MCC for ordinal classifiers. The use of Cohen's linear or quadratic weighted and unweighted κ -measures is found suitable for assessing the performance of multi-class classifiers [8]. However, Czodrowski [8] do not distinguish ordinal classifiers. Cohen's unweighted κ -measure is compared to

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a bunch of metrics, including the F1 score in the accuracy of assessing the performance of binary and multi-class classifiers for balanced and imbalanced data compositions by Ferri et al. [5], without specifically accounting for ordinal classifiers. It is observed that the unweighted κ -measure shows similar performance as the metric called accuracy for multi-class classifiers in general. However, it shows similar performance with the F1 score for large datasets with more than 1000 observations [5]. Serious concerns have also been raised against the use of Cohen's κ -measure in assessing the performance of multi-class classifiers. There is a strong correlation between Cohen's unweighted κ -measure and MCC, and high κ values are observed for poorly performing classifiers with imbalanced data while MCC was insensitive to this case [12]. Specifically, mean absolute error (MAE), its variations, and mean squared error (MSE) are considered for the evaluation of ordinal classification performance under imbalanced data compositions [13,14]. Cardoso and Sousa [15] propose an ordinal classification index (OCI), consider Misclassification Error Rate (MER) for ordinal classification performance, and compare MER with MAE and MSE. However, they do not consider mainstream and (weighted) agreement metrics for ordinal classification performance.

In image classification studies, Cohen's κ -measure is found useful for assessing multi-class classifiers' performance. Some performance comparisons among well-known metrics, F1 score and Cohen's κ are reported. The grounds for using kappa-like agreement measures to assess multi-class classifiers' performance is that the confusion matrix is essentially a cross table showing the agreement between two raters, represented by the observations and estimations. In this case, the confusion matrix is taken as an "agreement table" that shows the classification of two raters into multiple classes of an outcome variable. Then, the level of agreement between two raters is equivalent to the level of goodness of classification. In this sense, all kappa-like agreement measures can be considered as an evaluation metric for ordinal classifiers. However, Cohen's κ -measure is proposed to be used when there is no ordering among the classes of the variable of interest (nominal type classes). Thus, it does not consider the degree of deviance from the main diagonal of the classification. For example, in classifying objects in images into the classes "bicycle", "car", "airplane", three classes are of nominal type since there is no hierarchy among them. When there is a hierarchy among the classes (ordinal type classes), we need to consider this hierarchy in the analysis. For example, diabetic retinopathy is a serious disease that may lead to visual impairment and can be prevented if detected early. For the detection, lesions related to the disease are screened in retinal images, and a classification is done based on the severity of stages using automatic image classification techniques [4]. Due to the natural hierarchy among the severity stages, we deal with an ordinal multi-class classification problem. In this case, the weighted version of Cohen's κ -measure, which includes the impact of the distance from the main diagonal, is available for use. Since Cohen's (weighted) κ -measure has some drawbacks [16,17], other measures such as Gwet's AC2, Scott's π (π), Brennan-Prediger's BP, and Krippendorff's α (α) are proposed in the agreement studies [18]. However, they are not considered as an evaluation metric for ordinal classifiers in the literature. When we consider ordinal classes, there are other weighting schemes besides linear and quadratic weights, such as ordinal, radical, ratio, circular, and bipolar weights to be used with weighted versions of the agreement measures to account for the ordinality of the classes [18]. Due to the interaction between the assumptions behind agreement measures and weighting schemes, each weighted agreement measure would perform satisfactorily in measuring a classifier's performance under a different data composition. Therefore, the aims of this

research are to (i) explore the precision/usefulness of weighted agreement measures as evaluation metrics for ordinal classifiers, (ii) compare the versatility of agreement metrics and the mainstream metrics under challenging compositions of confusion matrices in different fields of practice for ordinal classification, and (iii) identify the promising metrics under practical circumstances and make recommendations about the best metric to use for the evaluation of ordinal classifiers.

To fulfill the aims, our objectives are to (i) implement the weighted agreement measures for the evaluation of ordinal classifiers for balanced and imbalanced data compositions, (ii) conduct a numerical study with synthetic confusion matrices to see the quality of the metrics under contesting compositions of confusion matrices, and (iii) run a second numerical study with the outputs of ordinal classifiers with real data from a vast variety of the fields to assess the sensitivity of metrics to small differences between ordinal classifiers. Under these objectives, we consider weighted versions of κ , π , α , BP and AC2 metrics and compare their performance against the mainstream metrics accuracy, recall, precision, F1, MCC and the recently proposed informational agreement through two extensive numerical studies. In total, we consider 37 metrics and assess them under artificially created scenarios composed of the true performance of the classifier, the degree of imbalance in the data (the composition of data), and different misclassification scenarios for the classes of the ordinal dependent feature in the first numerical study. This numerical study reveals the metrics' performance/usefulness under various confusion matrix formations. In the second numerical study, we compare the metrics in terms of their sensitivity in distinguishing two classifiers with similar classification performance using 40 real datasets, including balanced, imbalanced, and extremely imbalanced data compositions from social science, life sciences, engineering, and other areas of practice. This numerical study is important to observe the ability of metrics to perceive even small differences in classification performance, which is a highly desired quality for an evaluation metric. The contributions of this study are that (i) we explore the performance of a wide range of unweighted and weighted agreement measures as metrics for ordinal multi-class classifiers, (ii) comparatively examine the performance of the mainstream metrics, and (iii) identify a metric that is superior to the mainstream metrics that can be used under different data compositions and areas of practice as a generic metric.

Section 2 outlines the related works in the literature. Section 3 describes the metrics considered in this study. Sections 4 and 5 present the numerical studies with synthetic confusion matrices and real data. Section 6 is devoted to the general recommendations and discussions.

2. Related works

The use of weighted κ and MCC as a loss function in deep learning models for image classification is considered by de La Torre et al. [4] for general image classification and Kook et al. [2] for the classification of complex data like images into ordinal classes. However, they only consider quadratic weights and Cohen's κ while there are better-performing alternatives. For hyperspectral image classification, both Deng et al. [1] and Selami and Tabbone [19] employ accuracy and unweighted κ to evaluate their proposed methods for multi-class classification.

Ben-David [20] focus on the expert systems for cost-sensitive applications and propose a new strategy based on weighted κ -measure to assess the performance of multi-class classifiers. They only consider unweighted, linear and quadratic weighted Cohen's κ -measure without distinguishing imbalance in data and comparing the accuracy of κ -measure to others such as accuracy, precision, recall, and F1 score. García et al. [21] consider

Cohen's unweighted κ -measure and classification rate for multi-class classifiers for genetics-based machine learning implementations. This study shows that the precision of metrics can vary for classification problems in different fields of application.

Ferri et al. [5] conduct an extensive experimental study to assess the reliability of 18 metrics in classification problems. They present very detailed background information about the different types of metrics, including their taxonomy. They only consider the unweighted version of κ -measure and do not include MCC in the comparison study. Some similarities between the F1 score and κ -measure, and accuracy and κ -measure are observed for different compositions of data by Ferri et al. [5]. Czodrowski [8] studies the advantages and disadvantages of Cohen's unweighted κ as a metric of performance for the classification problem in machine learning and cheminformatics. He creates various data compositions to compare the reaction of precision, recall, accuracy, κ , prevalence, and bias and observes that κ -measure is a useful metric within this set of metrics. Saito and Rehmsmeier [22] compare the performance of precision/recall plots to Receiver Operating Characteristics (ROC) plots in assessing the binary classifiers on imbalanced data. They consider the metrics computed using the information in the confusion matrix, such as accuracy, F1 score, MCC, and precision and observe that performance of precision/recall plots is superior to ROC plots. Korotcov et al. [23] compare the performance of deep neural networks to that of mainstream machine learning methods by using AUC, F1 score, Cohen's unweighted κ and MCC. They observe similar κ and MCC values for implementing deep learning and machine learning methods with pharmaceutical data. Since we work with the metrics relying on the qualitative understanding of error that take the number of false classifications into account [5] in this study, we do not consider either ROC or AUC. The metrics, such as logarithmic loss, require prediction probabilities. However, we directly work with the confusion matrices for better generalizability instead of working with a specific model. Therefore, metrics requiring prediction probabilities are not considered in this study.

For ordinal multi-class classification, it is essential to account for the severity of the error. In that sense, the use of metrics that do not count for the magnitude of the error, such as accuracy metric, is not suitable [13,15]. Gaudette and Japkowicz [13] consider MAE and MSE to capture the magnitude of the error and find that MAE and MSE perform better than the metric accuracy for imbalanced datasets. Baccianella et al. [14] propose macro-averaged versions of MAE for imbalanced datasets but do not compare them with the mainstream metrics. When the number of ordinal levels increases, it is appropriate to use MAE and MSE based on the approximation to the continuous scale. However, when the number of levels is not large enough to support such an approximation, it is essential to identify the promising metrics that capture the magnitude of error for ordinal classification. Cardoso and Sousa [15] define OCI directly using the confusion matrix and consider the relative ordering of true and predicted classes (concordant and discordant pairs) and their deviation from the main diagonal of the confusion matrix. The weighted agreement metrics considered in this study capture concordance and discordance in the confusion matrix and account for the magnitude of discrepancy between the observed and predicted classification of the labels, in addition to the degree of similarity between them. Thus, figuring out the usefulness and efficiency of the weighted agreement metrics in the evaluation of ordinal classification performance is an important contribution.

Boughorbel et al. [9] propose optimizing the metrics to handle the imbalance in data and develop a new binary classifier based on the optimization of MCC for imbalanced data. They compare MCC, AUC, Accuracy, and F1 under imbalanced data

composition and observe that AUC and MCC have consistent performance across different classifiers with imbalanced data composition. Rácz et al. [6] conduct a statistical analysis on the performance of 28 metrics, including unweighted κ , MCC, F1 score, accuracy, diagnostic odds ratio, and AUC by taking data composition (imbalanced/balanced), level of classification (binary/multi-class), and the performance of metric as factors against the sum of ranking differences over three datasets. They observe that most of the metrics are sensitive to the composition of the dataset and the F1 score and the diagnostic odds ratio are the least sensitive ones. Wardhani et al. [24] focus on imbalanced data composition and compare the reliability of F1 score, g-mean, MCC, unweighted κ , and AUC metrics over an empirical study on the cabbage image classification. They observe that accuracy, F1 score, g-mean, MCC, and κ provide similar results for different confusion matrices while AUC is sensitive to the changes. When the degree of imbalance is high, Wardhani et al. [24] suggest not to use MCC and unweighted κ -measure to avoid misleading results.

In recent studies, Casagrande et al. [25] propose a new metric called informational agreement (IA) to measure the strength between two assessors for binary and multi-class labels to avoid the disadvantages of κ -measure. They test IA under the cases where the κ -measure gives problematic results by taking two diagnostic classifiers as assessors. Delgado and Tibau [12] focus on the similarities and differences between Cohen's unweighted κ -measure and MCC under various compositions of the confusion matrix for binary and multi-class classifiers. They theoretically and numerically study the equivalence between MCC and unweighted κ and figure out the formations of confusion matrix where unweighted κ should be avoided. Chicco and Jurman [11] consider MCC and observe its advantages over the F1 score and accuracy for binary classification. They conclude that MCC is a more reliable metric for assessing performance for binary classifiers and should be preferred over accuracy and F1 score.

There is a close theoretical relationship between κ -measure and MCC [12,26] when the confusion matrix is symmetric. Delgado and Tibau [12] also observe that κ -measure differs from MCC and is not reliable under imbalanced compositions of data. This result motivates us to explore other alternatives of κ -measure to identify a better metric for assessing the performance of multi-class classifiers with imbalanced datasets. Different to the existing studies, we consider four alternatives of the κ -measure with six weighting schemes under different data compositions in this study.

In general, different studies in different areas have produced contradictory conclusions on the suitability of MCC and Cohen's κ -measure. Although the characteristics of different areas are influential in this contradiction, these studies are limited to their simulation or numerical experimentation spaces. This is another motivation for conducting extensive numerical studies with synthetic and real data from different fields to explore the precision/usefulness of other weighted agreement measures as metrics for multi-class classifiers for ordinal labels.

3. Performance metrics for multi-class classifiers

Evaluation metrics differ according to the way they handle the error. The measures based on the qualitative understanding of error take into account the number of false classifications. Those regarding the probabilistic understanding of error are based on the distance from the true probability and are mostly used in reliability studies. The metrics related to the ranking accuracy of the model are used when the classifiers are evaluated on how well they select a given number of best labels/subjects from a dataset such as recommender systems [5]. In this article, we focus on the metrics relying on the qualitative understanding of error.

Table 1

The confusion matrix for a classifier.

		Predicted class				Row margin
		1	2	...	R	
Actual class	1	n_{11}	n_{12}	...	n_{1R}	$n_{1.}$
	2	n_{21}	n_{22}	...	n_{2R}	$n_{2.}$
	...	...	...	...	...	...
	R	n_{R1}	n_{R2}	...	n_{RR}	$n_{R.}$
	Column margin	$n_{.1}$	$n_{.2}$	...	$n_{.R}$	n

3.1. Mainstream metrics

The confusion matrix for a multi-class classifier assigning n labels into R classes is shown in Table 1. The rows of the confusion matrix represent the actual classes, and the columns show the predicted classes. In Table 1, n_{ij} denotes the number of labels that are actually in class i and predicted to be in class j , where $i, j = 1, 2, \dots, R$. The corresponding cell probability is $p_{ij} = n_{ij}/n$. The row and column totals are shown as row and column margins, respectively. Marginal row and column probabilities are $p_{i.} = n_{i.}/n$ and $p_{.j} = n_{.j}/n$, respectively. Since the row-totals of the confusion matrix are fixed by the frequencies in the dataset, the sampling scheme we are working with is the product-multinomial sampling [27].

Using the confusion matrix in Table 1, the mainstream metrics considered in this article are calculated as follows:

- *Accuracy* is the most common metric defined as the ratio of correctly predicted labels to the total number of labels.

$$Acc = \frac{\sum_{i=1}^R n_{ii}}{n}. \quad (1)$$

- *Macro-average recall* is also called balanced accuracy and computed as the arithmetic mean of recalls for all classes.

$$M.Recall = \frac{\sum_{i=1}^R recall_i}{R}, \quad \text{and} \quad recall_i = \sum_{j=1}^R n_{ji}/n_{i.}. \quad (2)$$

The macro-average recall is called “recall” in the rest of the article.

- *Macro-average precision* is defined as the arithmetic mean of precisions over R classes.

$$M.precision = \frac{\sum_{i=1}^R precision_i}{R}, \quad \text{and} \quad precision_i = \sum_{j=1}^R n_{ij}/n_{i.}. \quad (3)$$

Macro-average precision is called “precision” in the rest of the article.

- *Mean F1-score* is calculated as the arithmetic mean of F1-scores over R classes.

$$MeanF1 = \frac{\sum_{i=1}^R F1_i}{R}, \quad \text{and} \quad F1_i = 2 \times \left(\frac{precision_i \times recall_i}{precision_i + recall_i} \right). \quad (4)$$

- *Macro F1-score* is computed as the harmonic mean of macro-averaged precision and macro-averaged recall, defined in Eqs. (2) and (3), respectively.

$$M.F1 = 2 \times \left(\frac{M.precision \times M.recall}{M.precision + M.recall} \right). \quad (5)$$

- *Matthews correlation coefficient* [9] given in Eq. (6) measures the degree of correlation between the predicted and actual

class memberships of labels using the cell counts of the confusion matrix. It takes values in $[-1, 1]$ where -1 represents the poorest, and 1 shows perfect classification performance:

$$MCC = \frac{n \sum_{i=1}^R n_{ii} - \sum_{i=1}^R n_{i.} n_{.i}}{\sqrt{(n^2 - \sum_{i=1}^R n_{i.}^2)(n^2 - \sum_{i=1}^R n_{.i}^2)}}. \quad (6)$$

- *Informational agreement* has recently been proposed by Casagrande et al. [25] and is based on the amount of information (entropy) exchanged between the raters in the agreement context. In the classification performance assessment, the higher the agreement between a classifier and the actual distribution of labels, the higher the classification performance of the classifier. It is computed as

$$IA = \frac{MI(X, Y)}{\min\{H(X), H(Y)\}}, \quad MI(X, Y) = \sum_{i,j=1}^R p_{ij} \log_R \left(\frac{p_{ij}}{p_{i.} p_{.j}} \right), \quad (7)$$

$$H(X) = - \sum_{i=1}^R p_{i.} \log_R(p_{i.}), \quad \text{and} \quad H(Y) = - \sum_{i=1}^R p_{.i} \log_R(p_{.i}).$$

3.2. Weighted agreement coefficients

The weighted agreement coefficients are essentially used to evaluate the level of agreement between two raters who classified the subjects into ordered categories. The general form for agreement coefficients (A) is defined as follows:

$$A = \frac{P_o - P_e(A)}{1 - P_e(A)}, \quad P_o = \sum_{i,j=1}^R w_{ij} p_{ij}, \quad (8)$$

where P_o is the observed agreement, P_e is the proportion agreement expected by chance, w_{ij} shows the weight assigned to cell (i, j) of the confusion matrix, and P_e depends on the agreement coefficient and is calculated as given in Table 2.

The general formulation of Krippendorff's α measure is as follows [18]:

$$\alpha_w = \frac{\left(1 - \frac{1}{n\bar{r}}\right) p_{a0} + \frac{1}{n\bar{r}} - \sum_{k,l=1}^R w_{kl} \pi_k \pi_l}{1 - \sum_{k,l=1}^R w_{kl} \pi_k \pi_l}, \quad (9)$$

where

$$p_{a0} = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^R \frac{r_{ik}(\bar{r}_{ik} - 1)}{\bar{r}(r_i - 1)}, \quad \bar{r}_{ik} = \sum_{l=1}^R w_{kl} r_{il}, \quad \text{and} \quad \pi_k = \frac{1}{n} \sum_{i=1}^n r_{ik} \bar{r}. \quad (10)$$

While computing Krippendorff's α for a confusion matrix, $r_{ik} = 2$ and $r_i = \bar{r} = 2 \cdot R$ in Eqs. (9) and (10).

The critical distinction between agreement metrics is the assumption about the marginal distributions of assessors, which corresponds to the confusion matrix's margins in the classification performance context. One of the margins of the confusion matrix is fixed by the total class frequencies in the data. Therefore, the agreement coefficients calculated assuming that one of the margins is fixed are expected to perform better in the classification performance evaluation. However, none of the agreement coefficients has this assumption straightforwardly. Scott's π_w assumes the homogeneity of margins [28]; hence, it is expected to be a precise metric when the margins of the confusion matrix get closer. Krippendorff's α does not require the margins of the confusion matrix to be homogeneous. It instead counts where the

Table 2
The calculation of the proportion agreement expected by chance.

Coefficient	Symbol	P_e
Weighted kappa	κ_w	$P_e = \sum_{i,j=1}^R w_{ij} p_i p_j$
Scott's pi	π_w	$P_e = \sum_{i,j=1}^R w_{ij} p_i p_j, \quad p_i = \frac{(p_i + p_i)}{2}$
Gwet	AC2	$P_e = \frac{1}{R(R-1)} \left(\sum_{i,j=1}^R w_{ij} \right) \sum_{i=1}^R p_i (1 - p_i), \quad p_i = \frac{(p_i + p_i)}{2}$
Brennan–Prediger	BP_w	$P_e = \frac{1}{R^2} \sum_{i,j=1}^R w_{ij}$

classifier matches with the data in classifying the labels. Therefore, α is expected to be highly sensitive to misclassifications. Brennan–Prediger's BP assumes that the marginal distributions of the confusion matrix are uniform. Thus, BP is expected to work well with balanced data compositions. Since Gwet's AC2 adjusts uniformity regarding the variation in each class, it can handle the unbalanced data compositions to produce precise results [29].

The weighting schemes (w_{ij}) considered in this article are computed as given in Eqs. (11)–(15), where $w_{ij} = 1$ when $i = j$ for $i, j = 1, 2, \dots, R$ [18].

- Unweighted: $w_{ij} = 1$.
- Linear weights:

$$w_{ij} = 1 - \frac{|i - j|}{R - 1}. \quad (11)$$

- Quadratic weights:

$$w_{ij} = 1 - \frac{(i - j)^2}{(R - 1)^2}. \quad (12)$$

- Ordinal weights:

$$w_{ij} = 1 - \frac{M_{ij}}{M_{\max}}, \quad M_{ij} = \max(i, j) - \min(i, j) + 1 \quad \text{and} \\ M_{\max} = \max(M_{ij}). \quad (13)$$

- Radical weights:

$$w_{ij} = 1 - \frac{\sqrt{|i - j|}}{\sqrt{R - 1}}. \quad (14)$$

- Bipolar weights:

$$w_{ij} = 1 - \frac{(i - j)^2}{M(i + j - 2)(2R - i - j)}, \quad (15)$$

where M is the maximum value of $\left(\frac{(i-j)^2}{(i+j-2)(2R-i-j)} \right)$ over the pairs of (i, j) .

Linear weights are proportional to the degree of misclassification made by the classifier. If a classifier assigns a label farther away from its true class, the penalty applied by linear weights increases. The degree of penalty due to misclassification increases quadratically with the quadratic weights, while the radical weights penalize misclassification at a degree between the linear and quadratic weights. Since the ordinal weights account for the ranks of assignments done by the classifier, only the ranks are reflected in the weights. Bipolar weights produce a similar

pattern as linear weights but with higher values of weights. They get closer to quadratic weights toward the end of scale [18]. In terms of the relationship between Cohen's κ -measure with no weight, linear, and quadratic weights, Warrens [26] shows that "Cohen's unweighted $\kappa <$ Cohen's linear weighted $\kappa <$ Cohen's quadratic weighted κ ". The selection of weights is discussed in the later sections.

To implement the agreement coefficients as evaluation metrics, any agreement coefficient in Table 2 is calculated with one of the weighting schemes described in Eqs. (11)–(15) using the counts in the confusion table given in Table 1. Since this calculation is straightforward and does not require any iterative algorithms, there is no difference between the computational cost of the agreement and the mainstream evaluation metrics.

4. Numerical experiments with synthetic confusion matrices

4.1. Experiment space and data generation

In this numerical study, we create synthetic confusion matrices to examine the behavior of the evaluation metrics mentioned in Section 3 against different formations of the confusion matrix for a classification task into three ordinal classes. We generate confusion matrices with different characteristics for a given true classification performance, calculate evaluation metrics, and compare evaluation metrics with the true classification performance. The generation of confusion matrices, independent of a specific model and classifier, allows us to examine a wide range of confusion matrix formations that can be observed in real practice; hence, it provides sufficient generalizability.

The generated scenarios include high (0.8), moderate (0.5), and low (0.2) levels of true accuracy for a classifier which also translate into high, moderate, and low performance in practice. Here, the true accuracy/performance (TA) is defined as the ratio of correctly classified labels to the number of labels in each class in the data (either test or training set). Therefore, the metric called accuracy is expected to give similar values to TA subject to some random variation from replicating the experiments. In this sense, accuracy will be used as a control metric to assess if the data generation approach is sufficient to generate the desired levels of accuracy, and the metric accuracy is not compared to other metrics.

Since one of the main factors impacting the metrics is the balancedness of the distribution of labels into the target classes, balanced, imbalanced, and extremely imbalanced structures are created in combination with the true accuracy. This factor is called the structure of the table (ST) in the rest of the manuscript. Based on the classifier's performance and the area of interest, the accuracy of classification may differ across the ordinal classes. To

account for this, the following cases are created by using different p_{ij} settings:

1. Labels belonging to *all 3 classes* are classified with an accuracy close to the true accuracy.
2. Labels belonging to *2 out of 3 classes* are classified with an accuracy close to the true accuracy, and the last one is classified with low accuracy.
3. Labels belonging to *only one of 3 classes* are classified with an accuracy close to the true accuracy, and the remaining ones are classified with low accuracy.

This factor is named “Case” in the numerical study and denoted as Case 1, 2, and 3 throughout the manuscript. Case 1 is the best scenario where the classifier performs equally well for all classes. Thus, we expect a metric value close to the given true accuracy. In Cases 2 and 3, the classifier fails to work sufficiently for at least one of the three classes. So, even for high true accuracy, the classifier's performance is poor for Case 2 and poorer for Case 3. Since the classifier does not perform well, we expect lower values than the given true accuracy in Cases 2 and 3.

The confusion matrices are generated using the product-multinomial sampling scheme for the sample sizes of 50, 100, 200, and 500 for each combination of ST, TA, and Case. Each scenario is replicated 1000 times. The p_{ij} values are created to reflect the ordinal classes, and the combinations of ST, TA, and Case are used as inputs of the `rTable.RxC()` function of `rTableICC` package of R software [27]. We arbitrarily distort the given true accuracy in Cases 2 and 3 to create situations that can occur in real practice. The resulting synthetic confusion matrices are shown in Table A.1 of Appendix A. To generate ordinal classes, we follow the approach explained by Tran et al. [17, see p. 996–997].

We use mean absolute error (MAE) and root mean squared error (RMSE) given in Eq. (16) to compare the metrics under each simulation scenario.

$$MAE = \frac{1}{r} \sum_{i=1}^r |X - \hat{X}_i| \quad \text{and} \quad RMSE = \sqrt{\frac{1}{r} \sum_{i=1}^r (X - \hat{X}_i)^2}, \quad (16)$$

where r is the number of replications, X is the true value of accuracy, and $\hat{X}_i$ is the performance or weighted agreement metric estimation in the i th replication.

4.2. Results

The mainstream metrics' RMSE and MAE values for all sample sizes, balanced, imbalanced, and extremely imbalanced data compositions are given in Table A.2, A.4, ..., A.12 of Appendix A. Those for the weighted agreement metrics are given in Table A.3, A.5, ..., A.13 of Appendix A. In this section, “mainstream metrics” refer to the metrics described in Section 3.1 except accuracy. Since we use the definition of accuracy to generate confusion matrices, we exclude it from the inferences in this section. For a full picture of comparisons between the mainstream and the weighted agreement metrics, the reader should refer to pairs of Table A.2 and A.3, A.4 and A.5, and so on. The inferences in this section are also made by considering these pairs of tables.

The results in Appendix A show that accuracy is very close to TA, as expected in all the scenarios considered. When the data composition is balanced, mainstream metrics except for MCC and IA and most agreement metrics produce accurate measurements of classifier performance for Case 1 for all levels of TA and sample sizes. MCC and IA produce large MAEs and RMSEs, in general. As the classifier's performance becomes heterogeneous across the classes (moving from Case 1 to 2 and 3) and TA reduces to lower levels (the classification task gets harder), mainstream metrics

get worse in accurately capturing true accuracy. For high and moderate true accuracy, Cohen's κ and Scott's π with ordinal weights measure the classification performance accurately for all cases. When the true accuracy is low, unweighted or quadratic weighted AC2 performs better than other measures for all cases. We observe that mainstream metrics have a slightly better or similar performance for balanced situations with high true accuracy than ordinal weighted Cohen's κ and Scott's π measures. However, if the true accuracy is at lower levels and the classification accuracy is not distributed evenly as in Cases 2 and 3, ordinal weighted Cohen's κ and Scott's π and quadratic weighted AC2 perform better for balanced data compositions. Following the theoretical results of Delgado and Tibau [12], Cohen's unweighted κ produces similar values to MCC in all the scenarios for balanced data composition. There are small deviations between Cohen's unweighted κ and MCC for Cases 2 and 3. As expected, MAE and RMSE values get slightly smaller as the sample size increases for all metrics.

The results for imbalanced data compositions are generally similar to those observed for balanced datasets. MCC and IA produce high error values for high and moderate levels of TA. Mainstream metrics perform well when the classifier's performance is homogeneous across the ordinal levels and the true accuracy is high or moderate. The mean F1 is the best-performing mainstream classifier. Cohen's κ and Scott's π with quadratic weights have very similar MAE and RMSE values to mainstream metrics except for MCC and IA for these scenarios. However, when the TA is reduced, and the classifier's performance is better for some classes and worse for the other classes (Cases 2 and 3), quadratic or ordinal weighted Cohen's κ and Scott's π produce a promising performance that is better than mainstream metrics for most of the scenarios. For low TA under imbalanced data composition, AC2, BP, and Krippendorff's α perform better than the mainstream and other agreement metrics when only one of the classes is captured with an accuracy close to 0.2. We observe slight decreases in MAE and RMSE for increasing sample sizes with imbalanced data composition.

When the data composition is extremely imbalanced, MAE and RMSE values of the mainstream metrics considerably increase if the classifier does not perform equally well in all classes. For heterogeneous classifier performance across the classes, quadratic weighted Cohen's κ and Scott's π generate better performance estimates than mainstream metrics for all sample sizes. For lower TA, the unweighted AC2 is the best metric among the agreement metrics for all cases. It performs better than the mainstream metrics for sample sizes greater than 50.

4.3. Conclusions

Among the mainstream metrics, MCC and IA produce notably larger MAE and RMSE values than most of the metrics in most scenarios. The mainstream metrics, except for MCC and IA, successfully identify the performance of classifiers when the data composition is balanced, and all three ordinal classes are captured equally successfully. For the scenarios that deviate from these ideal conditions, error margins increase for the mainstream metrics.

For the agreement measures, we observe the theoretical relationship between MCC and Cohen's κ demonstrated by [12,26] in practice through the synthetic datasets when all the classes are assigned with similar success by the classifier and the data composition is balanced. However, this theoretical relationship is not observed under the deviations from this well-balanced scenario.

For the rest of the weighted agreement measures, a wide range of different values are observed, implying that agreement

measures are sensitive to different data compositions and true accuracy levels and have the potential to be used as reliable, useful metrics for multi-class classifiers. Specifically, Cohen's κ and Scott's π with quadratic weights show the best MAE and RMSE performance for high and moderate true agreement, data composition scenarios, and classification cases. Since Scott's π assumes the homogeneity of margins [28], it returns low error values when the margins of the confusion matrix get closer (the true agreement is moderate or high under all cases). However, when the true agreement is low, we deviate from this case; hence, other agreement measures perform better. When the true agreement is low, and data composition is imbalanced or extremely imbalanced, we observe that AC2 gives low MAE and RMSE values with different weights. The reason for Gwet's AC2 to perform better is that AC2 adjusts uniformity due to the variation created by the low true agreement and imbalance in the data [29].

All the considered mainstream and agreement metrics are sensitive to the sample size. We get smaller MAE and RMSE values as the sample increases. The benefit of increasing sample size on MAE and RMSE becomes notable when the true accuracy is high or moderate with balanced data composition. When the imbalance increases and one or two of the classes are correctly classified (Case 2 and 3), increasing the sample size does not help reduce the values of MAE and RMSE by a considerable margin.

Overall, when the data composition is imbalanced for the synthetic confusion matrices, the agreement measures mostly with ordinal weights produce lower error measures than the mainstream metrics and IA for most cases. They are more sensitive than MCC against unsuccessful classification in at least one of the classes. However, we need to study the response of agreement metrics to the composition of data in a generalizable setting to identify which agreement measure and weighting scheme combination can be used for performance assessment of ordinal multi-class classifiers and how their usage is compared to that of the mainstream metrics, MCC and the recently introduced metric IA. For this purpose, we conduct an extensive numerical study with real datasets in the next section.

5. Numerical experiments with real data

5.1. Datasets and classifiers

In the numerical study with real data, the sensitivity of 37 metrics of Section 3 is assessed in distinguishing ordinal classifiers. Since there is no unique metric to be used as the gold standard for determining the superiority of one metric over another, we focus on the sensitivity of metrics to the slight difference between the performance of two classifiers. It is a desired quality for a metric to distinguish subtle differences in the classification performance.

We use 40 real datasets having dependent features with multiple ordinal classes. For the generalizability of the results, we gather datasets from the main areas of practice: social sciences (16), life sciences (13), engineering (4) and other areas (7). Datasets and their web links are given in Table B.1 of Appendix B. Summary information about the datasets is given in Table B.2 of Appendix B. The sample size of each dataset is given under the "# Labels" column of Table B.2. For the datasets having more than 5000 labels, a sub-sample of size 1000 is taken randomly. Data composition for each dataset is qualified based on the balancedness of marginal probabilities. Overall, we have 6 balanced, 8 imbalanced, and 26 extremely imbalanced datasets. We have sufficient coverage of the number of features ranging from 3 to 56 and the number of ordinal classes from 3 to 11 among the datasets. There are very small samples as well as very large samples from a wide range of areas of practice included in this numerical study.

We aim to create 10 folds of data from each dataset if it has more than 100 labels. Since the confusion matrix becomes very sparse for the datasets with less than 100 labels, we create 2 folds for lsci2, lsci4, lsci13, eng1, other2, and other7 datasets. We run the computations separately for the full dataset, balanced, imbalanced, extremely imbalanced, life sciences, social science, engineering, and other areas datasets for each fold. Since we cannot create 10 folds for every dataset, we have a different number of replications for each run composed of datasets and their folds, as given in Table 3. For example, for the run with "All datasets", we have 10 folds for 34 datasets and 2 folds for 6 datasets; in total, we run 352 replications for all datasets. For the run with "Only ENG" datasets, there are 10 folds for 3 out of 4 datasets and 2 folds for 1 dataset, resulting in 32 replications. We utilize the runs for the folds as replications to conduct statistical hypothesis tests.

In the implementation, we employ pairs of classifiers developed for ordinal data. The classifiers are selected by considering that they have similar overall classification performances. We assess if the metrics are able to differentiate between the classifiers. The difference between the actual performances of classifiers is an important consideration in assessing the precision/usefulness of the metrics. Conceivably, in real practice, the precision of metrics is not a very important issue if one of the classifiers is performing far better than the other. The success of a metric lies in its ability to distinguish the classifiers when their performances are close to each other. Therefore, to create two classifiers with similar performance, we use two classifiers from the same classification method with slight changes in their parameter settings. We consider support vector machines with ordered partitions (SVMOP), which is developed for classification into ordinal classes by Waegeman et al. [30], weighted K-nearest neighbors for ordinal classification (WKNNOR) proposed by Hechenbichler and Schliep [31], and kernel discriminant learning for ordinal regression (KDLOR) proposed by Sun et al. [32]. There can be other choices of ordinal classifiers. However, since we are not comparing the performance of actual classifiers/classification methods and need to get just confusion matrices out of classifiers, the choice of SVMOP, WKNNOR, and KDLOR classifiers does not pose any problem with the generalizability of the results on the performance of the metrics.

The parameters γ and *cost* impact the performance of SVMOP classifiers, which use the Gaussian radial basis kernel function. The γ parameter adjusts the scale of the Gaussian kernel. A large scale γ for a given *cost* creates wide classification boundaries. For smaller scales, under-fitting occurs if classification boundaries become more focused under limited regions of the space [33, see p. 347–348]. An increased under-fitting reduces the classification performance with the test set. We set the *cost* value for both SVMOPs to 1 and use the γ parameter to create classifiers that have similar overall performance but make mistakes on ordinal classes. We set the SVMOP classifiers to use with $\gamma = 1$ and 10 with *cost* = 1 for SVMOP₁ and SVMOP₂, respectively.

For WKNNOR classifiers, we use the Euclidean distance and Gaussian kernel for both classifiers and set the number of neighbors differently to create two classifiers with slightly different performances. A KNN with a small number of neighbors tends to over-fit, but a large value of the number of neighbors has the potential to create under-fit [33, see p. 160 and 352]. We set the number of neighbors to 20 for WKNNOR₁ and 5 for WKNNOR₂ to create a small discrepancy between the performances of the classifiers that we expect the evaluation metrics to distinguish.

For KDLOR classifiers, we use the Gaussian kernel with different scales to create KDLOR₁ and KDLOR₂ classifiers. We set the scale of the Gaussian kernel to 1 and 5 for KDLOR₁ and KDLOR₂, respectively. Since KDLOR is a kernel-based method,

Table 3
Runs, datasets, and the corresponding number of replications.

Runs with	Number of datasets	Dataset number	Replications
All datasets	40	1 to 40	352
Only balanced	6	3, 9, 12, 17, 20, 39	52
Only imbalanced	8	2, 4, 5, 6, 11, 14, 38, 40	72
Only extremely imbalanced	26	1, 7, 8, 10, 13, 15, 16, 18, 19, 21 to 37	228
Only SSCI	16	14 to 29	160
Only ENG	4	30 to 33	32
Only LSCI	13	1 to 13	106
Only OTHER	7	34 to 40	138

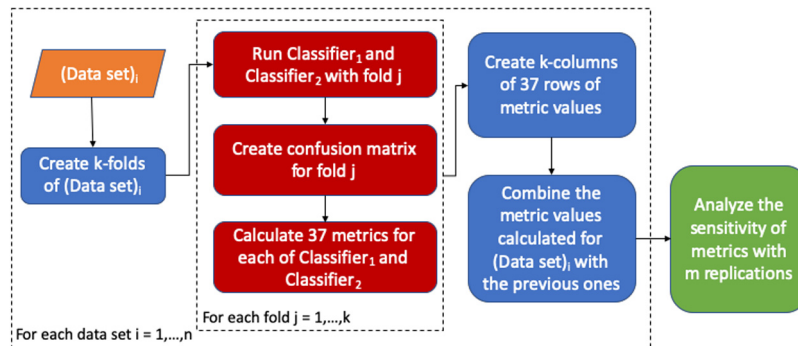


Fig. 1. Flow diagram of the numerical study with real data. Here, n shows the number of datasets in each particular run as given in the second column of Table 3, $k = 2, 10$, and the values of m are shown in the last column of Table 3 for each run.

its performance is characterized by the kernel. The scale of the Gaussian kernel specifies the boundaries of the regions created for classification, similar to SVMOPs.

It should be noted that these settings do not ensure that either Classifier₁ or Classifier₂ under each ordinal classification method will have superior performance. The superiority issue is outside the focus of this numerical study since we only need to have one of the classifiers perform slightly differently.

We employ `svmofit()`, `wknnor()`, and `kdlortrain()` functions from **OCAPIS** R package [34] for the implementation of SVMOP, WKNNOR, and KDLOR classifiers, respectively. The flow diagram of implementation is given in Fig. 1. We create k -folds, $k = 2, 10$ of each dataset and implement k -fold cross-validation without aggregating the results coming out of the folds.

5.2. Results

Results are composed of the values of 37 metrics observed over 2 sub-classifiers under each of 3 main classification methods, 40 real datasets, and replications from k -folds. The real datasets are also divided into categories based on their balancedness and the field of application. We follow Demšar [35] to provide statistical evidence on the significance of the difference between metrics across the factors used for comparisons. We use multi-way analysis of variance (ANOVA) and Tukey's pairwise comparison test to identify the metrics that produce significantly different values between classifiers under each main classification method. In this way, we identify the sensitive metrics to separate the classification performance of two similar classifiers from each other. The normality and homogeneity of variances assumptions of ANOVA [35] need to be ensured to get valid inferences from ANOVA. For this aim, the Kolmogorov–Smirnov normality test and Leneve test for the homogeneity of variances are implemented using `notest` [36] and `car` [37] R packages, respectively.

5.2.1. Visual inspection

The means of metrics, calculated by aggregating over 40 datasets and folds, are given in Figs. 2, 3, and 4 for SVMOP, WKNNOR, and KDLOR classifiers, respectively, along with the $\pm$ standard error limits. Note that the statistical significance of differences is thoroughly discussed in the parts following these figures. In Figs. 2, 3, and 4, we see different behaviors for each group of metrics, namely, mainstream, Cohen's κ , Scott's π , AC2, BP and Krippendorff's α for all the ordinal classification methods. There is no clear distinction between the values of metrics with different weights. For all the main classification methods, SVMOP, WKNNOR, and KDLOR, almost all metrics show that Classifier₁ has slightly better performance than Classifier₂.

For SVMOP classifiers (Fig. 2), weighted π metrics have notably different behavior than all the other metrics. Weighted κ and α metrics show similar results, while weighted AC2 and BP metrics indicate a similar level of performance to each other. However, AC2 and BP metrics do not distinguish SVMOP₁ and SVMOP₂, which is not the desired result. In this sense, weighted π metrics show the highest difference between the classifiers. Among the mainstream metrics, macro recall, macro precision, and macro F1 are close to each other, while IA and accuracy show a similar performance. MCC is the only mainstream metric that shows a slight difference between the classifiers. Overall, weighted π metrics are able to distinguish the classifiers as desired, while none of the mainstream metrics is capable of for the SVMOP method.

For WKNNOR classifiers (Fig. 3), weighted AC2 and BP indicate considerably higher performance for both classifiers than the other metrics. The first classifier is found to be slightly better by all the agreement measures. Since the mean metric values have the lowest standard errors for the WKNNOR classifiers, distinguishing between two classifiers becomes even more challenging. The mainstream metrics, except accuracy and IA, are insensitive to the difference between WKNNOR₁ and WKNNOR₂. BP is the most sensitive metric to the difference between two WKNNOR classifiers. Weights have no impact on the α metric and little

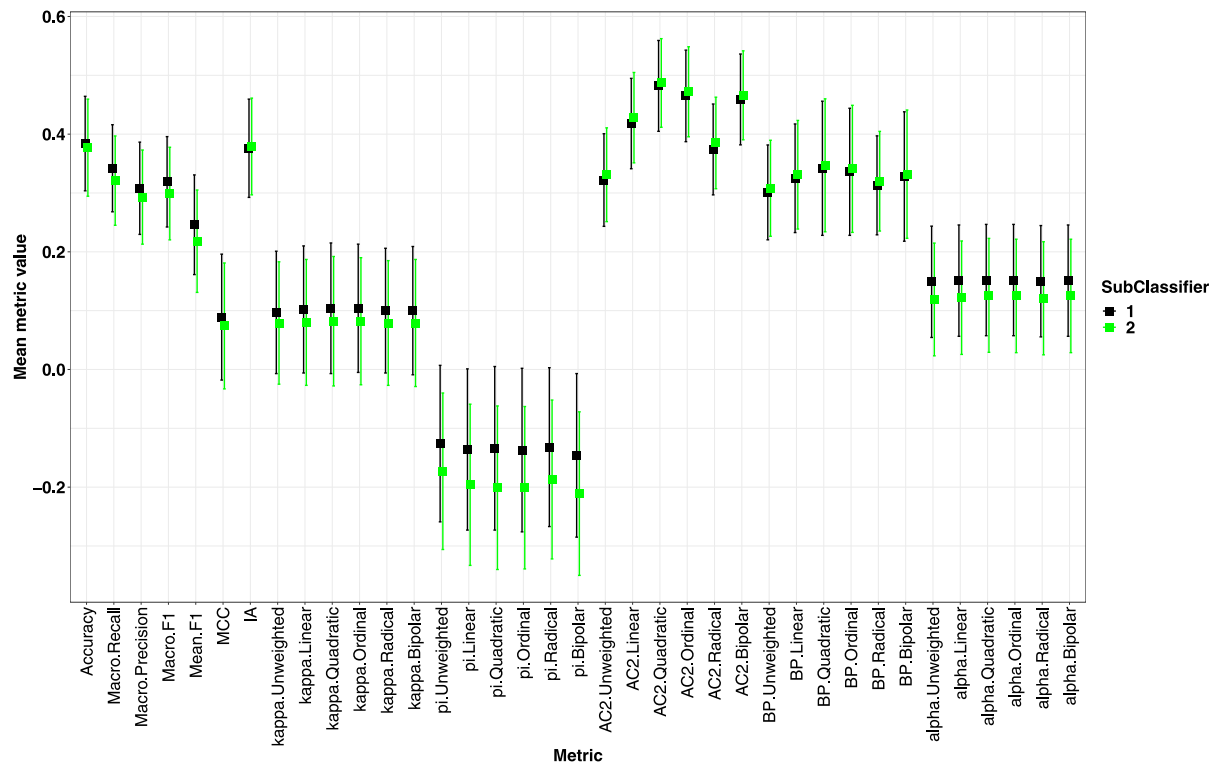


Fig. 2. Mean of metrics for the SVMOP classifiers. The mean is calculated over 40 datasets and folds. Bars show the limits of mean $\pm$ standard error.

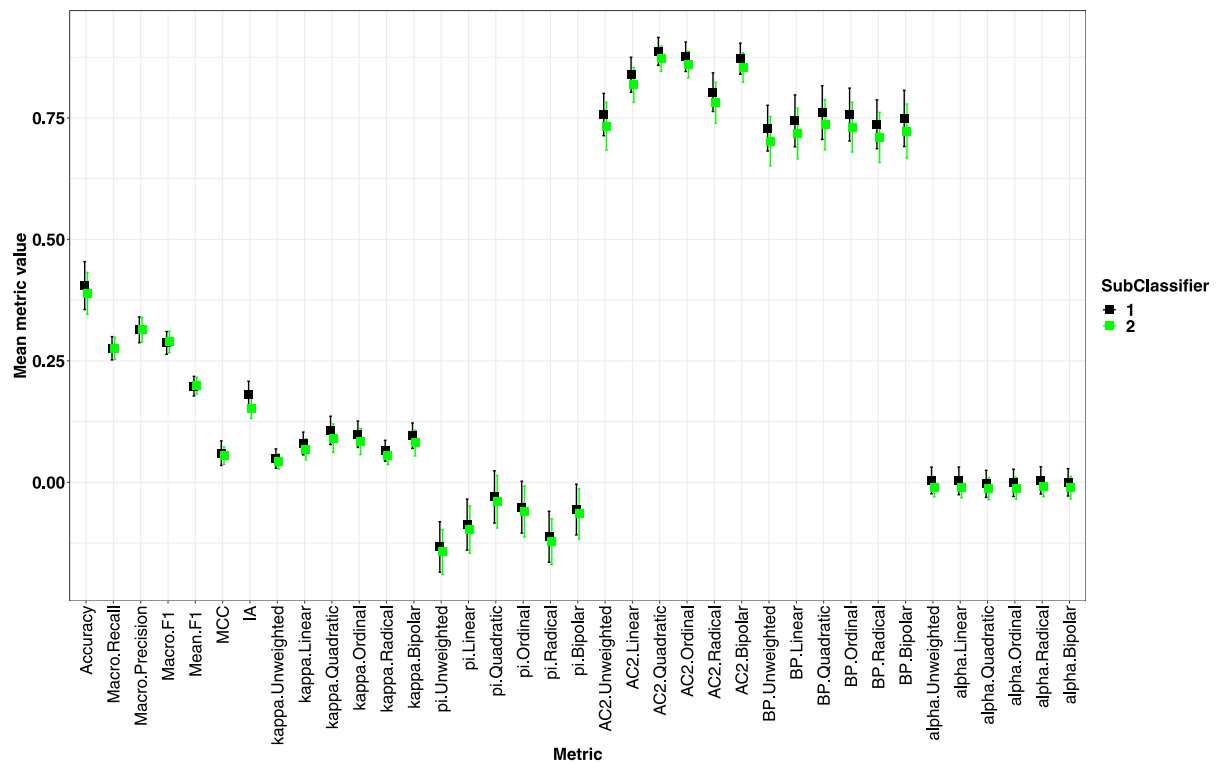


Fig. 3. Mean of metrics for the WKNNOR classifiers. The mean is calculated over 40 datasets and folds. Bars show the limits of mean $\pm$ standard error.

impact on κ and BP metrics. Overall, while most mainstream metrics are insensitive to the slight difference between WKNNOR

classifiers, all agreement measures show some degree of difference between the performances of WKNNOR₁ and WKNNOR₂. BP

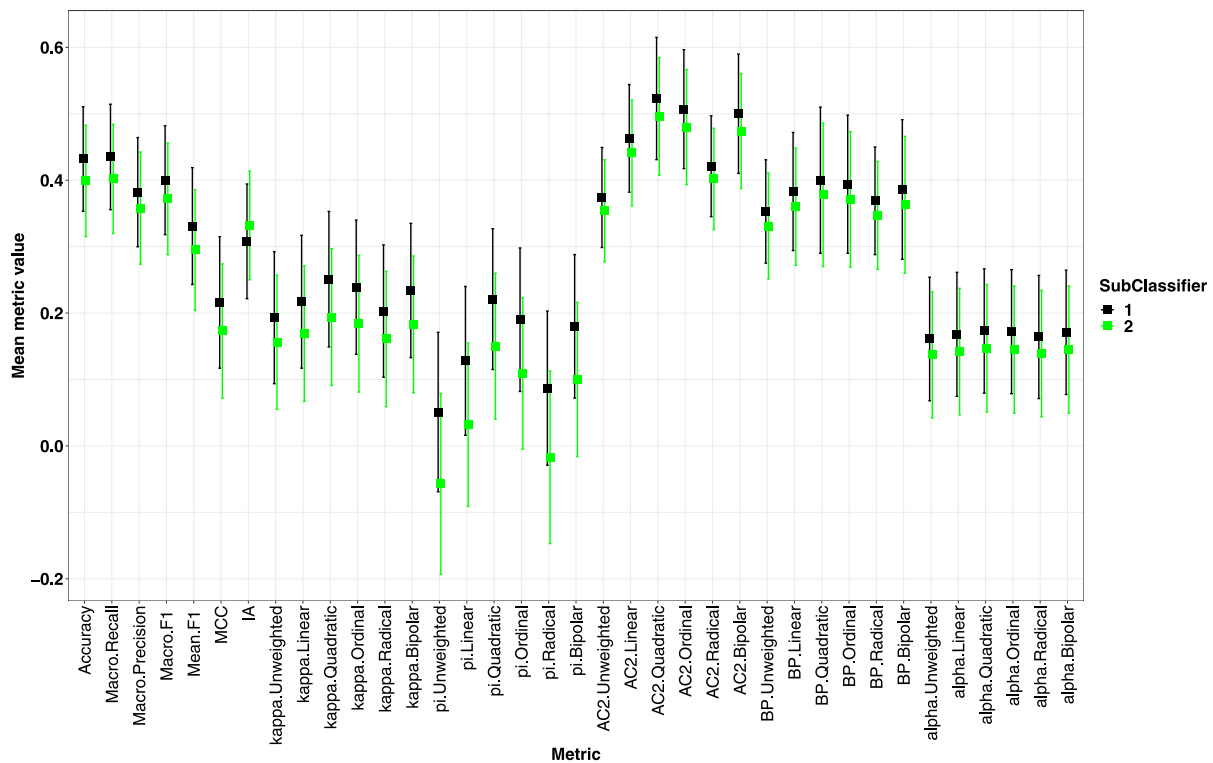


Fig. 4. Mean of metrics for the KDLOr classifiers. The mean is calculated over 40 datasets and folds. Bars show the limits of mean $\pm$ standard error.

shows the best sensitivity when the metric values are very close to each other (very small standard errors).

For KDLOr classifiers (Fig. 4), we observe higher standard errors among the values of metrics that increase the sensitivity of the metrics to the slight difference between KDLOr₁ and KDLOr₂ classifiers. All the metrics are sensitive to the difference between the classifiers. The α metric is not impacted by the weights for KDLOr classifiers too. The π metrics show the highest sensitivity, and the weights do not influence the degree of their sensitivity. Although MCC has the highest sensitivity among the mainstream metrics, it is considerably lower than the difference indicated by the π metric. Overall, MCC, κ , and π metrics show similar levels of performance, while π distinguishes KDLOr₁ and KDLOr₂ as desired.

When the standard errors of metric values are high, the π metric is able to distinguish classifiers with similar performance as seen for SVMOP and KDLOr methods. However, when the metric values are less variable, BP indicates the difference between WKNNOR classifiers better than π . Regarding the level of difference captured by the metrics, the weighting scheme does not have a notable influence on both π and BP metrics. However, it impacts the classification performance indicated by these metrics.

Since the data composition and the field of application are two important features of the datasets, we need to consider these factors for the inferences. Figs. 5 and 6 show the mean metric values and their standard error limits aggregated over all the classifiers and the ordinal classification methods. From these plots, we observe that both the data composition and the application field considerably impact the mean metric values generated for the classifiers since the metrics react differently to each of balanced, imbalanced, and extremely imbalanced data compositions and datasets from different fields. Note that since the aggregation is done over very different ordinal classification methods for Figs. 5 and 6, it is not appropriate to interpret the differences between the data compositions and application fields without looking into the data composition for each ordinal

classification method and the application field for each ordinal classification method. However, Figs. 5 and 6 demonstrate that we need to consider each data composition for each ordinal classification method.

5.2.2. Statistical tests

In order to give detailed significance test results for the difference between two sub-classifiers, we first apply ANOVA with metrics and sub-classifiers, ensure the normality and homogeneity of variances assumptions of ANOVA, and then apply Tukey's pairwise comparison tests to identify the metrics for which the sub-classifiers are giving different results. The results of these tests are given in Table B.3 of Appendix B. Suppose we refer to Tukey's pairwise comparison test result for the metric MCC. The test indicates whether there is a significant difference between MCC values computed for Classifier₁ and Classifier₂. If there is a significant difference, we can conclude that MCC is able to distinguish the performance difference between Classifier₁ and Classifier₂ as desired. In this way, we gather statistical evidence to identify the metrics that perform as desired in distinguishing two classifiers with similar performance. If a pairwise comparison of two metrics is insignificant, we conclude that both metrics are unsuccessful in distinguishing classifiers; hence, they are insensitive to the difference between them.

For SVMOP classifiers, the p-values of Tukey's pairwise comparison tests are given in Fig. 7 in the breakdown of data compositions. For balanced datasets, pairwise comparisons of all mainstream metrics except MCC and IA are insignificant, implying that they are insensitive to the difference between SVMOP₁ and SVMOP₂ classifiers. All weighted κ and π metrics and some weighted AC2 metrics are significantly different from the mainstream metrics, while BP and some α metrics are not. All the weighted π metrics are significantly different from the other mainstream and agreement metrics. When the composition of data becomes imbalanced, mean F1, MCC, and IA metrics give

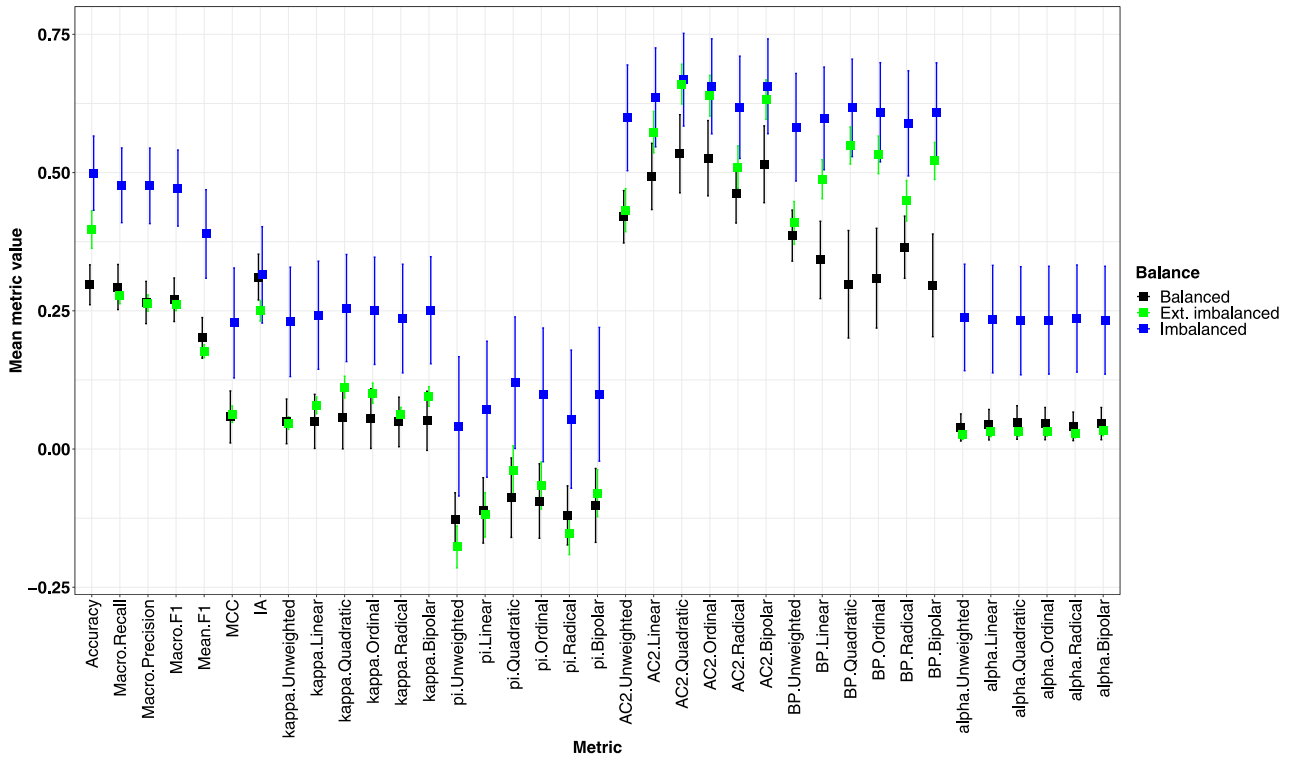


Fig. 5. Mean of metrics according to the data composition. The mean is calculated over all classifiers and folds. Bars show the limits of mean $\pm$ standard error.

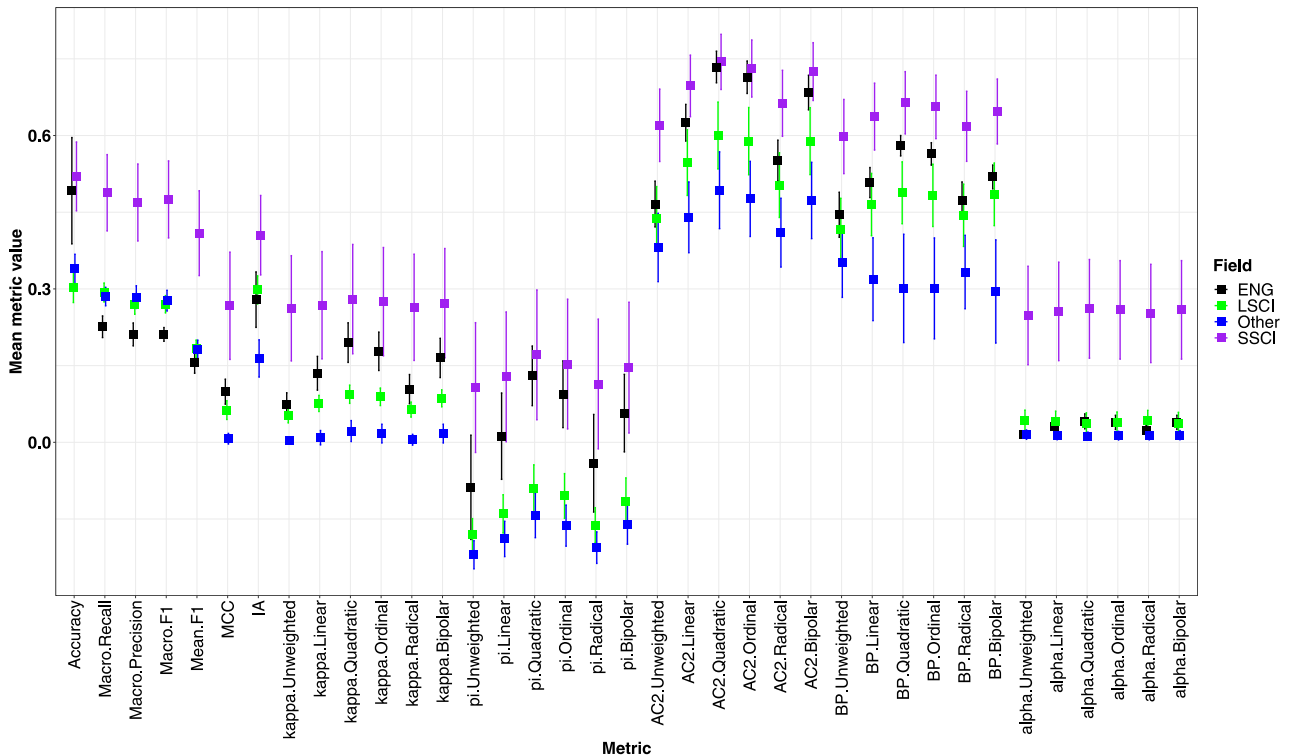


Fig. 6. Mean of metrics according to the application fields of the datasets. The mean is calculated over all classifiers and folds. Bars show the limits of mean $\pm$ standard error.

significantly different values between SVMOP₁ and SVMOP₂ classifiers among the mainstream metrics. κ and α metrics are not significantly different from MCC, while all other agreement metrics are significantly different from MCC and all other mainstream metrics. For extremely imbalanced datasets, in addition to the

inferences given for imbalanced data composition, we see mean F1 becomes significantly different from other mainstream and agreement metrics.

The p-values of Tukey's pairwise comparison tests for metrics are given in Fig. 8 in the breakdown of the field of application

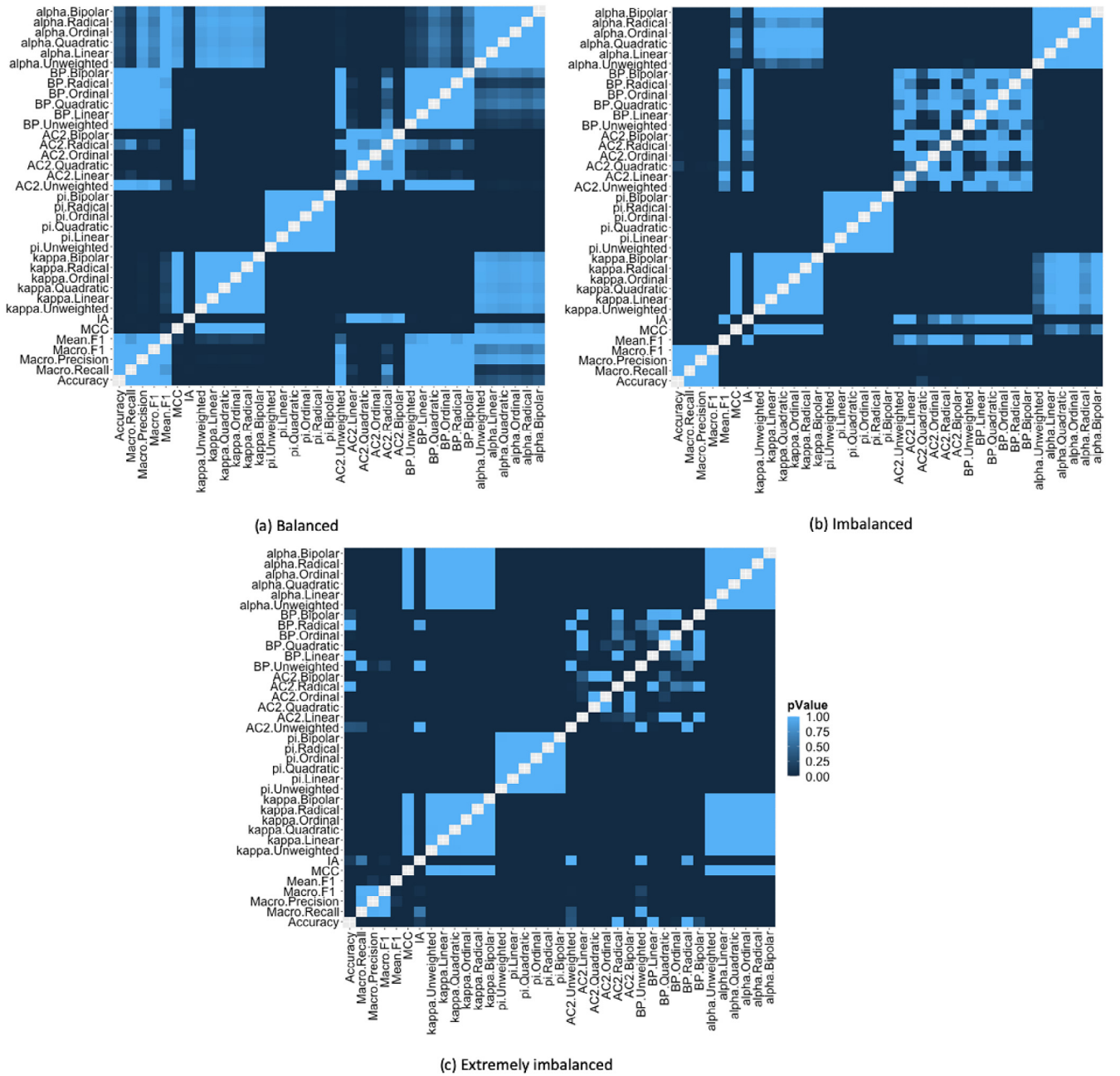


Fig. 7. The p-values of Tukey's pairwise comparison tests for SVMOP classifiers in the breakdown of data composition. The axes of the plots show the metrics calculated for SVMOP₁ and SVMOP₂ classifiers.

for SVMOP classifiers. We get different patterns of the pairwise comparisons' p-values across the application fields; hence, the application field makes a difference in the usefulness of the metrics. Among the mainstream metrics, only MCC is significantly different across two SVMOP classifiers for social sciences datasets. Since all κ , π , and α metrics are significantly different from the mainstream metrics, their behavior against two similar SVMOP classifiers differs. Also, there are significant differences between all weighted κ and π metrics. All weighted α metrics are not significantly different from κ but show the difference for the rest of the agreement measures. Some weighted AC2 and BP metrics also show a similar result. We observe very similar results for life sciences datasets. In addition to MCC, the only difference is that mean F1 and IA metrics also show significant differences among the mainstream metrics. The results of datasets categorized in other application fields are similar to life sciences, but

accuracy is also sensitive to the difference between SVMOP₁ and SVMOP₂ classifiers. For engineering datasets, the metric accuracy significantly differs from all other mainstream metrics, κ , π , and α metrics. There is no significant difference between accuracy and weighted AC2 and BP metrics. Consistently, weighted versions of AC2 and BP metrics are significantly different from the mainstream metrics except for accuracy.

For WKNNOR classifiers, the p-values of Tukey's pairwise comparison tests are given in Figure B.1 of Appendix B in the breakdown of data compositions and Figure B.3 of Appendix B in the breakdown of application fields. We get similar results for balanced and imbalanced datasets. For both data compositions, all the agreement metrics are significantly different from each other and mainstream metrics. Among the mainstream metrics, mean F1, MCC, and IA are sensitive to the differences between

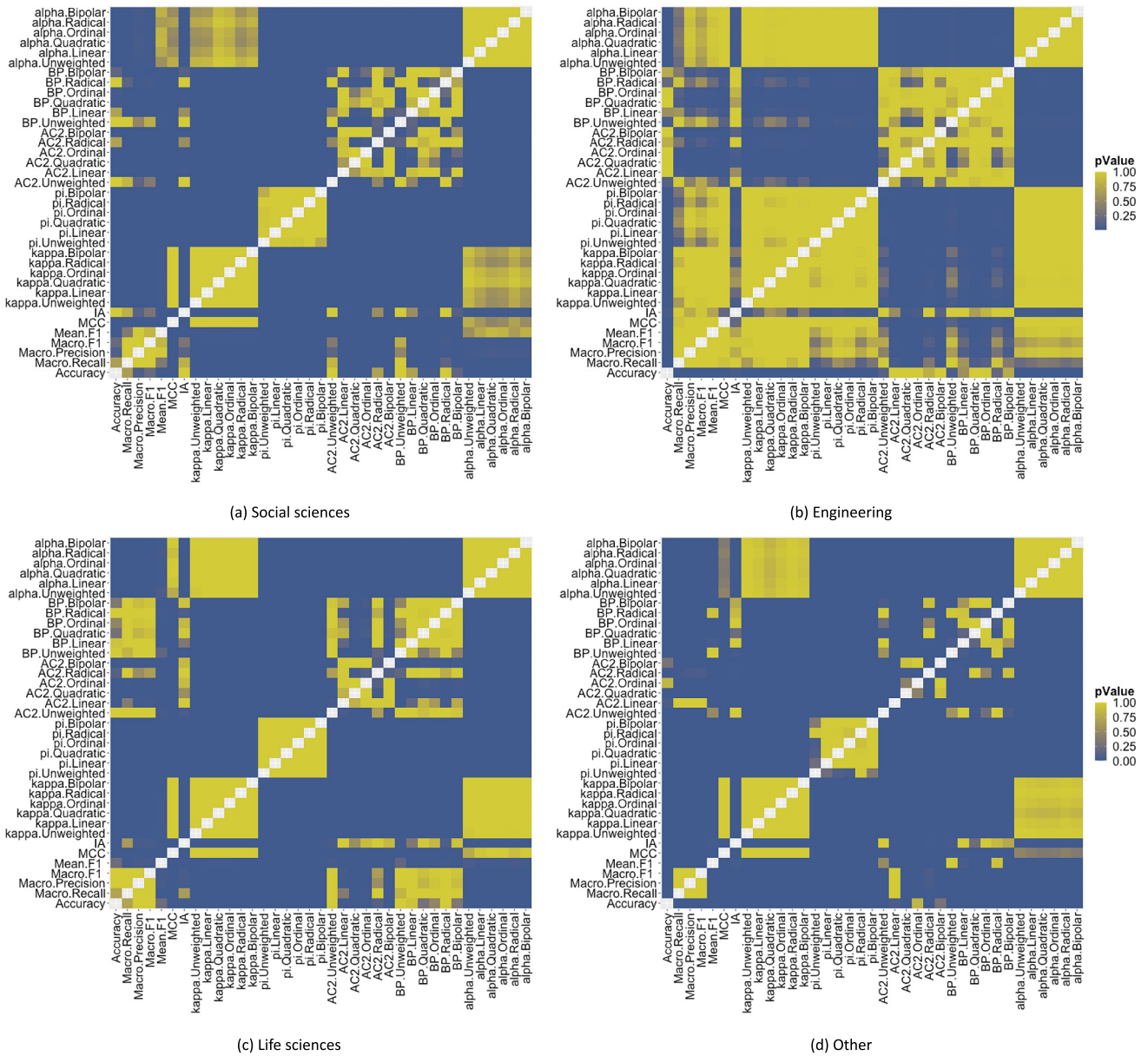


Fig. 8. The p-values of Tukey's pairwise comparison tests for SVMOP classifiers in the breakdown of the field of application. The axes of the plots show the metrics calculated for SVMOP₁ and SVMOP₂ classifiers.

WKNNOR₁ and WKNNOR₂ classifiers. Only accuracy shows a significant difference among the mainstream metrics for extremely imbalanced datasets. Some of κ , most of π , and all AC2, BP, and α metrics are sensitive to the difference between two WKNNOR classifiers. Scott's π metrics are not significantly different from κ and α metrics. We observe different patterns of p-values across the fields of application in Figure B.3 of Appendix B. For social science datasets, κ metrics are not significantly different from the mainstream metrics except for accuracy. Scott's π and Krippendorff's α metrics show sensitivity to the discrepancy between the classifiers compared to mainstream metrics. AC2 and BP metrics are significantly different from the mainstream and other agreement metrics. When it comes to engineering datasets, only accuracy, AC2, and BP metrics are significantly sensitive to the difference between WKNNOR₁ and WKNNOR₂ classifiers, and α shows significant differences from these metrics. Scott's π and Cohen's κ metrics are insensitive for engineering datasets. For life sciences and other datasets, we see similar patterns of p-values. All the metrics except macro recall and macro precision

are sensitive to the difference between WKNNOR₁ and WKNNOR₂ classifiers.

For KDLOR classifiers, the p-values of Tukey's tests are given in Figure B.2 and B.4 of Appendix B in terms of data compositions and application fields, respectively. For balanced datasets, MCC is the only sensitive metric to the difference between KDLOR₁ and KDLOR₂ classifiers. While κ , π , and α metrics are not significantly different from MCC, AC2 and BP metrics are different. π and α show significant differences from all other metrics except κ for balanced datasets. For imbalanced datasets, mean F1, MCC, IA, and all agreement metrics differ significantly from the mainstream metrics. The metric α shows different behavior than the other agreement metrics. When the data composition becomes extremely imbalanced, MCC, κ , π , and α metrics are sensitive to the difference between KDLOR classifiers, and the difference between this group of metrics and AC2 and BP is significant. For the application fields, we observe different patterns of p-values for social sciences, engineering, and life sciences datasets, while datasets from other application areas produce a similar pattern

as the life sciences datasets. MCC, π , and α metrics produce significantly different results for KDLOR₁ and KDLOR₂ classifiers for social sciences datasets. AC2 and BP metrics are significantly different from the rest of the agreement and mainstream metrics. For engineering datasets, interestingly, only unweighted and radical weighted versions of the π metrics show a significant difference for the KDLOR classifiers in addition to AC2 and BP metrics. None of the mainstream and κ metrics is sensitive to the difference between KDLOR₁ and KDLOR₂. Most metrics are sensitive to the difference between KDLOR classifiers for life sciences and other datasets. Only accuracy, macro recall and macro F1 are insensitive.

5.2.3. Impact of the number of features

To further investigate the sensitivity of the metrics against different numbers of features, we worked with datasets ssci15, ssci14, other3, eng2, ssci13, ssci11, lsci4, lsci5, lsci11 from Table B.2 of Appendix B. The numbers of features of ssci15 and ssci11 are reduced to 2 and 7, respectively, to have $f = 2, 3, \dots, 10$ features in the considered datasets. The procedure outlined in Fig. 1 is implemented with SVMOP, WKNNOR, and KDLOR classifiers for each dataset. The two-sample t-test is used to assess if there is a significant difference between Classifier₁ and Classifier₂. The p-values of the tests are given in Table B.4, B.5 and B.6 of Appendix B for SVMOP, WKNNOR, and KDLOR classifiers, respectively.

In Table B.4, B.5 and B.6 of Appendix B, cells with bold font show the metrics that produce a significant difference between the two classifiers; hence, they are sensitive to the slight difference between the classification performance of classifiers. For all classification methods of SVMOP, WKNNOR, and KDLOR, the sensitivity of all the metrics is not impacted by the increasing number of features. For SVMOP, the mainstream metrics are insensitive for all the considered number of features. Weighted κ and π metrics detect the difference just for the dataset with seven features. Metrics are more sensitive to the difference between WKNNOR classifiers. While the mainstream metrics are sensitive to the difference between WKNNOR classifiers with 6 features, weighted agreement metrics are sensitive for multiple numbers of features. The best sensitivity results are seen for the weighted BP metric, which detects the difference for 2, 5, 6, and 7 features. For KDLOR classifiers, the weighted κ and π metrics show sensitivity, with the datasets having 5 and 7 features.

5.3. Conclusions

Under different data compositions, Scott's π metrics are consistently sensitive to the difference between classifiers for all ordinal classification methods, while mainstream metrics, except MCC and IA are not, in general. Other agreement metrics are as consistent as Scott's π in distinguishing classifiers under different data compositions. Specifically, metrics have difficulty in separating the performance of classifiers for extremely imbalanced datasets where Scott's π metric works best. Among the agreement metrics, while κ , π , and α generally have similar behavior, AC2 and BP have similar responses to the slight difference between the classifiers. All the metrics perform better in distinguishing the classifiers for the SVMOP method and the worst performance for the KDLOR classifiers.

Engineering datasets produce notably different pairwise comparison results for all ordinal classification methods than the other fields of application. The results for the datasets categorized as other fields of application or life sciences are generally similar. Among the agreement metrics, π and κ metrics do not work as desired for engineering datasets, while AC2 and BP are mostly sensitive to the differences between the classifiers. For the mainstream metrics, only the accuracy metric is sensitive for

SVMOP and WKNNOR classifiers for engineering datasets. For all the other categories of datasets, κ , π , and α metrics are sensitive to the difference between classifiers. From the set of mainstream metrics, only MCC is consistently sensitive across different fields of application.

The mainstream metrics are generally insensitive to the difference between the classifiers, except for MCC. The mean F1 and MCC show sensitivity against the slight difference between the classifiers for some ordinal classifiers and application fields.

The number of features has no impact on the ability of both mainstream and weighted agreement metrics to distinguish the multi-class classification performance of classifiers. The sensitivity of the metrics is highly related to the characteristics of the datasets as investigated in Sections 5.2.1 and 5.2.2.

6. Discussion

Assessing the performance of classifiers in supervised machine/deep learning is crucial to choosing the classifier to employ. This problem gets challenging when the classification task is a multi-class classification with ordinal classes. In this work, we focused on the metrics used to assess the performance of ordinal multi-class classifiers based on the qualitative understanding of error. Cohen's κ -measure has been used in applications as one of the promising metrics for classifiers without distinguishing the type of multi-class categorical responses as ordinal or nominal. Many other agreement measures and their weighted versions for ordinal data are proposed in the literature. Furthermore, from the previous works, it is known that there are better performing measures than Cohen's κ in the area of agreement studies. Considering these, we conducted two extensive numerical studies with 37 metrics composed of the mainstream and agreement metrics to investigate the reliability/usefulness of the agreement measures as evaluation metrics for ordinal multi-class classifiers. We identified Scott's weighted π -measure as a strong alternative to Cohen's weighted κ .

In the first numerical study, the ability of metrics to capture the classification performance as close as the true accuracy is analyzed through randomly generated synthetic confusion matrices under different data compositions and different levels of accuracy in the classification performance. Ordinal multi-class classification processes are not perfect. Thus, they are prone to the composition of ordinal response and sample size. We show that the composition of data considerably influences the accuracy of metrics. The mainstream metrics, except MCC, are observed to be insensitive to capture different types of misclassification. In contrast, the metrics based on agreement measures react better to misclassification in individual classes of the dependent feature. Since the mainstream metrics have the main diagonal of the confusion matrix and one of its margins in their formulations, they do not capture the misclassification that occurs in the off-diagonal cells. Only MCC takes both margins into account and shows better sensitivity than the other mainstream metrics. On the other hand, (weighted) agreement metrics consider diagonal and off-diagonal cells and margins (row and column totals) of the confusion table to capture the correct classifications as well as the magnitude of divergence from the correct classifications. The level of true accuracy is another important factor on the performance of metrics. When the true accuracy is low, the metrics' margin of error to evaluate the classification performance increases, mainstream metrics become highly insensitive to misclassification, and the range of agreement measures increases. Therefore, the reliability of all metrics is higher when they indicate high performance. The case with a low true accuracy is challenging because it pushes the cell counts to off-diagonal cells of the confusion table. The metrics that do not incorporate off-diagonal cells of the confusion table

in their formulation got more negatively impacted in detecting a low level of accuracy. Cohen's κ and Scott's π with quadratic weights perform better than all the considered mainstream metrics and agreement measures under the most challenging data compositions.

The second numerical study uses 40 real datasets, including balanced, imbalanced, and extremely imbalanced data compositions from social sciences, life sciences, engineering, and other areas, to create replications via cross-validation. In this study, we examined the sensitivity of metrics to small differences between two similar classifiers from the same ordinal classification method. A useful metric is expected to be sensitive enough to distinguish such classifiers. We observed that MCC successfully discriminates two classifiers with similar performance. However, it generates high margins of error when the data composition is imbalanced or extremely imbalanced. Scott's quadratic weighted π and Cohen's quadratic weighted κ metrics show promising sensitivity for challenging cases, including extremely imbalanced data compositions.

The quadratic weights penalize misclassification quadratically; hence, the agreement metrics with quadratic weights are highly sensitive to misclassification. Scott's π does not have a uniformity assumption on the row and column totals of the confusion table. Instead, it assumes the homogeneity of margins. If the margins of a confusion table are considerably non-homogeneous, it implies that the classifier assigns objects to the wrong labels at a notably high rate. This translates into very poor performance, and the metrics can capture this case straightforwardly. Thus, in general, Scott's π is not impacted by the composition of data. Cohen's κ does not have restrictive assumptions on the margins of the confusion table as well. Therefore, Scott's quadratic weighted π and Cohen's κ generally show satisfactory preciseness. However, the assumptions, such as the uniformity of the margins of the confusion table BP or AC2, limits their preciseness for imbalanced data compositions. Due to the theoretical relationship between Cohen's κ and MCC for symmetric confusion matrices, MCC also show satisfactory preciseness when the confusion matrix is near-symmetric.

Based on the results of both numerical studies, Scott's π and Cohen's κ metrics with quadratic weights are both sensitive to the small differences between classifiers and produce close values to the true level of accuracy in most of the considered scenarios. Therefore, they are recommended to be used in practice, in general. Specifically for engineering applications and low true performance, the use of quadratic weighted AC2 metric is recommended. We recommend avoiding accuracy, recall, precision, and macro F1 for extremely imbalanced datasets. Mean F1 has a better performance than macro F1 but is not as good as Scott's weighted π . In some cases, the recently proposed metric, IA, is useful in the performance assessment of ordinal classifiers. However, it is not recommended for general use.

This study focuses on a performance analysis of the evaluation metrics readily computed using a given confusion matrix. The computation of all the evaluation metrics requires straightforward analytical calculations without any iterative method, done in milliseconds without consuming a noticeable computer memory. Therefore, the space and time complexity of calculating evaluation metrics is outside the focus of our study. The main limitations of our study include the simulation space of synthetic data study and the number of datasets and replications of the real-data study. Although we cover many scenarios of true accuracy, data composition, misclassification, and application areas, there can still be other application-specific characteristics to influence the performance of metrics.

In addition to the theoretical results on the similarity of unweighted Cohen's κ and MCC, we numerically observed that

Scott's unweighted π produces close values to MCC under some scenarios. Besides, Gwet's unweighted AC2, Brennan–Prediger's unweighted BP, and Krippendorff's α metrics produce very close values to IA when the true agreement is high and, in some cases of low and moderate agreement. A theoretical investigation of these results is a future direction.

Another future direction is using Scott's quadratic weighted π as a loss function for ordinal image classification and regression. Cohen's weighted κ showed improved generalization ability for image classification [4] when used as a loss function. In this sense, the use of Scott's quadratic weighted π as a loss function would provide further improvement.

CRediT authorship contribution statement

Ayfer Ezgi Yilmaz: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Review. **Haydar Demirhan:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.asoc.2023.110020>. Tables A.1 to A.14, Tables B.15 to B.17 and Figures B.1 to B.3 are given in the Supplementary Material.

References

- [1] C. Deng, X. Liu, C. Li, D. Tao, Active multi-kernel domain adaptation for hyperspectral image classification, *Pattern Recognit.* 77 (2018) 306–315.
- [2] L. Kook, L. Herzog, T. Hothorn, O. Dürr, B. Sick, Deep and interpretable regression models for ordinal outcomes, *Pattern Recognit.* 122 (2022) 108263.
- [3] L. Li, L. Ma, L. Jiao, F. Liu, Q. Sun, J. Zhao, Complex contourlet-CNN for polarimetric SAR image classification, *Pattern Recognit.* 100 (2020) 107110.
- [4] J. de La Torre, D. Puig, A. Valls, Weighted kappa loss function for multi-class classification of ordinal data in deep learning, *Pattern Recognit. Lett.* 105 (2018) 144–154.
- [5] C. Ferri, J. Hernández-Orallo, R. Modroiu, An experimental comparison of performance measures for classification, *Pattern Recognit. Lett.* 30 (1) (2009) 27–38.
- [6] A. Rácz, D. Bajusz, K. Héberger, Multi-level comparison of machine learning classifiers and their performance metrics, *Molecules* 24 (15) (2019) 2811.
- [7] Y. Sasaki, R. Fellow, The Truth of the F-Measure, University of Manchester: MIB-School of Computer Science, 2007.
- [8] P. Czodrowski, Count on kappa, *J. Comput. Aided Mol. Des.* 28 (11) (2014) 1049–1055.
- [9] S. Boughorbel, F. Jarray, M. El-Anbari, Optimal classifier for imbalanced data using Matthews Correlation Coefficient metric, *PLoS One* 12 (6) (2017) e0177678.
- [10] D. Chicco, M.J. Warrens, G. Jurman, The Matthews correlation coefficient (MCC) is more informative than Cohen's Kappa and Brier score in binary classification assessment, *IEEE Access* (2021).

- [11] D. Chicco, G. Jurman, The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation, *BMC Genomics* 21 (1) (2020) 6.
- [12] R. Delgado, X.-A. Tibau, Why Cohen's Kappa should be avoided as performance measure in classification, *PLoS One* 14 (9) (2019) e0222916.
- [13] L. Gaudette, N. Japkowicz, Evaluation methods for ordinal classification, in: Y. Gao, N. Japkowicz (Eds.), *Advances in Artificial Intelligence*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2009, pp. 207–210.
- [14] S. Baccianella, A. Esuli, F. Sebastiani, Evaluation measures for ordinal regression, in: 2009 Ninth International Conference on Intelligent Systems Design and Applications, IEEE, 2009, pp. 283–287.
- [15] J.S. Cardoso, R. Sousa, Measuring the performance of ordinal classification, *Int. J. Pattern Recognit. Artif. Intell.* 25 (08) (2011) 1173–1195.
- [16] A.E. Yilmaz, T. Saracbası, Assessing agreement between raters from the point of coefficients and log-linear models, *J. Data Sci.* 15 (1) (2017) 1–24.
- [17] D. Tran, A. Dolgun, H. Demirhan, Weighted inter-rater agreement measures for ordinal outcomes, *Comm. Statist. Simulation Comput.* 49 (4) (2020) 989–1003.
- [18] K.L. Gwet, *Handbook of Inter-Rater Reliability: The Definitive Guide To Measuring the Extent of Agreement Among Raters*, Advanced Analytics, LLC, 2014.
- [19] A. Sellami, S. Tabbone, Deep neural networks-based relevant latent representation learning for hyperspectral image classification, *Pattern Recognit.* 121 (2022) 108224.
- [20] A. Ben-David, Comparison of classification accuracy using Cohen's Weighted Kappa, *Expert Syst. Appl.* 34 (2) (2008) 825–832.
- [21] S. García, A. Fernández, J. Luengo, F. Herrera, A study of statistical techniques and performance measures for genetics-based machine learning: accuracy and interpretability, *Soft Comput.* 13 (10) (2009) 959.
- [22] T. Saito, M. Rehmsmeier, The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets, *PLoS One* 10 (3) (2015) e0118432.
- [23] A. Korotcov, V. Tkachenko, D.P. Russo, S. Ekins, Comparison of deep learning with multiple machine learning methods and metrics using diverse drug discovery data sets, *Mol. Pharm.* 14 (12) (2017) 4462–4475.
- [24] N.W.S. Wardhani, M.Y. Rochayani, A. Iriany, A.D. Sulistyono, P. Lestantyo, Cross-validation metrics for evaluating classification performance on imbalanced data, in: 2019 International Conference on Computer, Control, Informatics and Its Applications (IC3INA), IEEE, 2019, pp. 14–18.
- [25] A. Casagrande, F. Fabris, R. Girometti, Beyond kappa: an informational index for diagnostic agreement in dichotomous and multivalued ordered-categorical ratings, *Med. Biol. Eng. Comput.* 58 (12) (2020) 3089–3099.
- [26] M.J. Warrens, A comparison of Cohen's kappa and agreement coefficients by Corrado Gini, *Int. J. Res. Rev. Appl. Sci.* 16 (2013) 345–351.
- [27] H. Demirhan, rTableICC: An R package for random generation of 2x2xK and RxC contingency tables, *R J.* 8 (1) (2016) 48–63.
- [28] R. Artstein, M. Poesio, Inter-coder agreement for computational linguistics, *Comput. Linguist.* 34 (4) (2008) 555–596.
- [29] K.L. Gwet, Computing inter-rater reliability and its variance in the presence of high agreement, *Br. J. Math. Stat. Psychol.* 61 (1) (2008) 29–48.
- [30] W. Waegeman, L. Boullart, et al., An ensemble of weighted support vector machines for ordinal regression, *Int. J. Comput. Syst. Sci. Eng.* 3 (1) (2009) 47–51.
- [31] K. Hechenbichler, K. Schliep, Weighted K-Nearest-Neighbor Techniques and Ordinal Classification, Collaborative Research Center 386, Discussion Paper 399, 2004, <http://dx.doi.org/10.5282/ubm/epub.1769>.
- [32] B.-Y. Sun, J. Li, D.D. Wu, X.-M. Zhang, W.-B. Li, Kernel discriminant learning for ordinal regression, *IEEE Trans. Knowl. Data Eng.* 22 (6) (2010) 906–910.
- [33] M. Kuhn, K. Johnson, et al., *Applied Predictive Modeling*, Vol. 26, Springer, 2013.
- [34] M.C. Heredia-Gómez, S. García, P.A. Gutiérrez, F. Herrera, Ocapis: R package for ordinal classification and preprocessing in scala, *Prog. Artif. Intell.* 8 (3) (2019) 287–292.
- [35] J. Demšar, Statistical comparisons of classifiers over multiple data sets, *J. Mach. Learn. Res.* 7 (2006) 1–30.
- [36] J. Gross, U. Ligges, nortest: Tests for normality, 2015, URL: <https://CRAN.R-project.org/package=nortest>, R package version 1.0-4.
- [37] J. Fox, S. Weisberg, *An R Companion To Applied Regression*, third ed., Sage, Thousand Oaks CA, 2019.